

Savitribai Phule Pune University

T.Y.B.Sc.(C.S.) (NEP – 2020) SEM – V Practical Examination

Lab Course CS-321-VSC-P

Foundation of Artificial Intelligence and Machine Learning

Duration: 3 Hours

Maximum Marks:35

Q1) Write a program or report that compares Minimax and Alpha-Beta Pruning on the same game tree. **[10 marks]**

Q2) Write a program to implement the Backward Chaining inference mechanism. **[20 marks]**

OR

Q2) Write a program to implement the Random Forest Classifier for classification tasks. Use the dataset provided: car_evaluation.csv (7 columns/features and 10 rows). All the features are of categorical data type, given by buying, maint, doors, persons, lug_boot, safety and class. Class is the target variable. Rename column names as ['buying', 'maint', 'doors', 'persons', 'lug_boot', 'safety', 'class']. Check the frequency counts of categorical variables. Check for missing values. Drop 'class' feature to create X and let y = 'class' variable. Split the dataset as 67:33% - train:test. Encode all the categorical variables. Implement the RF classifier with default parameters. Display the accuracy of the RF classifier. **[20 marks]**

No.	buying	maint	doors	persons	lug_boot	safety	class
1	vhigh	vhigh	2	2	small	low	unacc
2	vhigh	vhigh	2	2	small	med	unacc
3	vhigh	vhigh	2	2	small	high	unacc
4	vhigh	vhigh	2	2	med	low	unacc
5	vhigh	vhigh	2	2	med	med	unacc
6	vhigh	vhigh	2	2	med	high	unacc
7	vhigh	vhigh	2	2	big	low	unacc
8	vhigh	vhigh	2	2	big	med	unacc
9	vhigh	vhigh	2	2	big	high	unacc
10	vhigh	vhigh	2	4	small	low	unacc

Q3) Viva **[05 marks]**